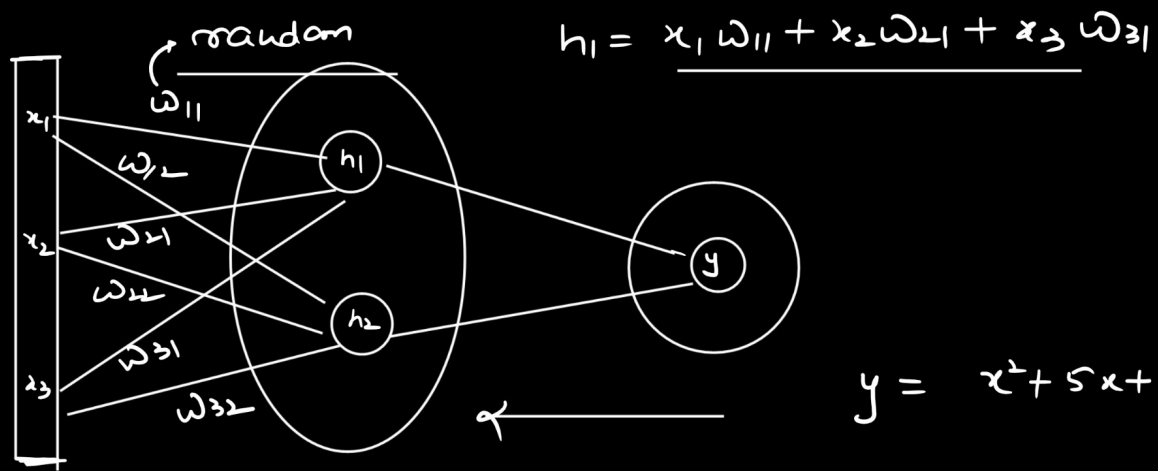


Pytorch

features	TensorFlow	Pytorch
Programming style	Keras API	Python & flexible
Training	model.fit()	Manual training loop
Automatic Differentiation	GradientTape()	Autograd
Model Building	Sequential/ -functions	nn.Module
Best suited for	Production & Deployment	Research & Experimentation

$$w_{new} = w_{old} - \eta * \frac{\delta L}{\delta w_{old}}$$

↗ gradient/derivation
↘ 0.001 ↘ backward()



$$y = x^2 + 5x + 2$$

y.backward() $\frac{\delta y}{\delta x} = 2x + 5$

$\omega_{new} = \omega_{old} - \eta * \frac{\delta L}{\delta \omega}$

$x = 3$

$2 * 3 + 5 = 11$

$\omega_{11}, \omega_{12}, \omega_{21}, \omega_{22} \dots \omega_{ij} \Rightarrow \underline{\underline{0}}$

channels $\rightarrow$ grayscale image
 $(1, 28, 28)$
 $\hookrightarrow$ image size

Model $\rightarrow$ (batch_size,

